COMPILER DESIGN

CSA1418

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Course	Compiler Design (CSA1418)

QUESTION

i) Construct LR(0) items, the state transition diagram, the parsing table, and perform parsing for the input string abab.

ii) Construct LR(1) items, the state transition diagram, the parsing table, and perform parsing for the input string abab.

Given grammar:

S → AA
A → aA | b

Augmented Grammar and Production Numbers

Production	**Rule**
(0)	S′ → S
(1)	S → AA
(2)	A → aA
(3)	A → b

Terminals: a, b, $ Non-terminals: S, A Input used for parsing: abab$

PART I — LR(0) PARSING

1. Canonical Collection of LR(0) Items

State	LR(0) Items
I_0	$S' \to \cdot S$ $S \to \cdot AA$ $A \to \cdot aA$ $A \to \cdot b$
$I_1 = GOTO(I_0, S)$	$S' \to S\cdot$
$I_2 = GOTO(I_0, A)$	$S \to A\cdot A$ $A \to \cdot aA$ $A \to \cdot b$
$I_3 = GOTO(I_0, a)$	$A \to a\cdot A$ $A \to \cdot aA$ $A \to \cdot b$
$I_4 = GOTO(I_0, b)$	$A \to b\cdot$
$I_5 = GOTO(I_2, A)$	$S \to AA\cdot$
$I_6 = GOTO(I_3, A)$	$A \to aA\cdot$

2. LR(0) State Transition Diagram

$I_0 \text{—}S\to I_1 \quad I_0 \text{—}A\to I_2 \quad I_0 \text{—}a\to I_3 \quad I_0 \text{—}b\to I_4$

$I_2 \text{—}A\to I_5 \quad I_2 \text{—}a\to I_3 \quad I_2 \text{—}b\to I_4$

$I_3 \text{—}A\to I_6 \quad I_3 \text{—}a\to I_3 \quad I_3 \text{—}b\to I_4$

From state	a	b	A	S
I_0	I_3	I_4	I_2	I_1
I_2	I_3	I_4	I_5	—
I_3	I_3	I_4	I_6	—

3. LR(0) Parsing Table

In LR(0), a completed item is reduced on every terminal, including the end marker $. Here r1: S → AA, r2: A → aA, and r3: A → b.

State	ACTION a	ACTION b	ACTION $	GOTO S	GOTO A
0	s3	s4	—	1	2
1	—	—	acc	—	—
2	s3	s4	—	—	5
3	s3	s4	—	—	6
4	r3	r3	r3	—	—
5	r1	r1	r1	—	—
6	r2	r2	r2	—	—

The table has no shift/reduce or reduce/reduce conflict; therefore, the grammar is LR(0).

4. LR(0) Parsing of abab$

Step	Stack	Remaining input	Action
1	0	abab$	s3
2	0 a 3	bab$	s4
3	0 a 3 b 4	ab$	r3: A → b; GOTO(3,A)=6
4	0 a 3 A 6	ab$	r2: A → aA; GOTO(0,A)=2
5	0 A 2	ab$	s3
6	0 A 2 a 3	b$	s4
7	0 A 2 a 3 b 4	$	r3: A → b; GOTO(3,A)=6
8	0 A 2 a 3 A 6	$	r2: A → aA; GOTO(2,A)=5
9	0 A 2 A 5	$	r1: S → AA; GOTO(0,S)=1
10	0 S 1	$	ACCEPT

Result: The input string abab is accepted by the LR(0) parser.

PART II — CANONICAL LR(1) PARSING

1. Canonical Collection of LR(1) Items

An LR(1) item is written as [production, lookahead]. Items with the same core and several lookaheads are combined using braces.

State	Canonical LR(1) Items
I_0	[S′ → ·S, $] [S → ·AA, $] [A → ·aA, {a,b}] [A → ·b, {a,b}]
I_1 = GOTO(I_0, S)	[S′ → S·, $]
I_2 = GOTO(I_0, A)	[S → A·A, $] [A → ·aA, $] [A → ·b, $]
I_3 = GOTO(I_0, a)	[A → a·A, {a,b}] [A → ·aA, {a,b}] [A → ·b, {a,b}]
I_4 = GOTO(I_0, b)	[A → b·, {a,b}]
I_5 = GOTO(I_2, A)	[S → AA·, $]
I_6 = GOTO(I_2, a)	[A → a·A, $] [A → ·aA, $] [A → ·b, $]
I_7 = GOTO(I_2, b)	[A → b·, $]
I_8 = GOTO(I_3, A)	[A → aA·, {a,b}]
I_9 = GOTO(I_6, A)	[A → aA·, $]

2. LR(1) State Transition Diagram

$$I_0 \xrightarrow{S} I_1 \quad I_0 \xrightarrow{A} I_2 \quad I_0 \xrightarrow{a} I_3 \quad I_0 \xrightarrow{b} I_4$$
$$I_2 \xrightarrow{A} I_5 \quad I_2 \xrightarrow{a} I_6 \quad I_2 \xrightarrow{b} I_7$$
$$I_3 \xrightarrow{A} I_8 \quad I_3 \xrightarrow{a} I_3 \quad I_3 \xrightarrow{b} I_4$$
$$I_6 \xrightarrow{A} I_9 \quad I_6 \xrightarrow{a} I_6 \quad I_6 \xrightarrow{b} I_7$$

From state	a	b	A	S
I_0	I_3	I_4	I_2	I_1
I_2	I_6	I_7	I_5	—
I_3	I_3	I_4	I_8	—
I_6	I_6	I_7	I_9	—

3. Canonical LR(1) Parsing Table

A reduction is entered only under the lookahead carried by the completed LR(1) item.

State	ACTION a	ACTION b	ACTION $	GOTO S	GOTO A
0	s3	s4	—	1	2
1	—	—	acc	—	—
2	s6	s7	—	—	5
3	s3	s4	—	—	8
4	r3	r3	—	—	—
5	—	—	r1	—	—
6	s6	s7	—	—	9

7	—	—	r3	—	—
8	r2	r2	—	—	—
9	—	—	r2	—	—

The canonical LR(1) table is conflict-free.

4. LR(1) Parsing of abab$

Step	Stack	Remaining input	Action
1	0	abab$	s3
2	0 a 3	bab$	s4
3	0 a 3 b 4	ab$	r3: A → b; GOTO(3,A)=8
4	0 a 3 A 8	ab$	r2: A → aA; GOTO(0,A)=2
5	0 A 2	ab$	s6
6	0 A 2 a 6	b$	s7
7	0 A 2 a 6 b 7	$	r3: A → b; GOTO(6,A)=9
8	0 A 2 a 6 A 9	$	r2: A → aA; GOTO(2,A)=5
9	0 A 2 A 5	$	r1: S → AA; GOTO(0,S)=1
10	0 S 1	$	ACCEPT

Result: The input string abab is accepted by the canonical LR(1) parser.

FINAL CONCLUSION

For the grammar S → AA and A → aA | b, both the LR(0) and canonical LR(1) parsing tables are conflict-free. The input string abab is successfully reduced to S and accepted by both parsers.